Linux Laboratory

Part 1 & Part 2

Course: GIK2NV – Data storage and management technologies

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Video presentation: uploaded separately on Canvas

Introduction

In this lab we act as Linux administrators for a company. The company has three new departments: **Engineering**, **Sales** and **HR**.

- In **Part 1** we create folders, groups, users and permissions for the departments, one command at a time.
- In **Part 2** we write a Bash script that does the same kind of work automatically, for any user and group name.

Environment: Ubuntu Linux. All commands are run as the `root` user (on a normal account, put `sudo` in front of each command).

Names we used:

Department	Group	Admin	Users
Engineering	Engineering	eng_admin	eng_user1, eng_user2
Sales	Sales	sales_admin	sales_user1, sales_user2
HR	HR	hr_admin	hr_user1, hr_user2

Part 1 – User management

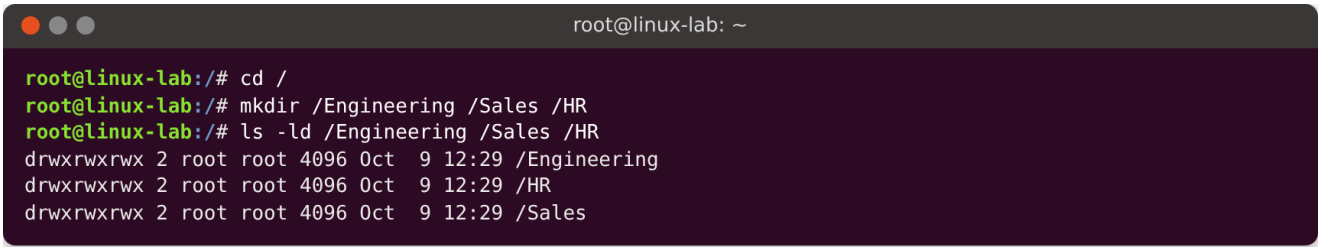
Task 1: Create a directory at the root (/) for each department

How we did it: We go to the root of the file system with `cd /`. Then we create one folder per department with `mkdir`. Each folder has the same name as its department.

Code:

```
cd /
mkdir /Engineering /Sales /HR
```

Verification: `ls -ld` shows that the three folders exist under `/`.



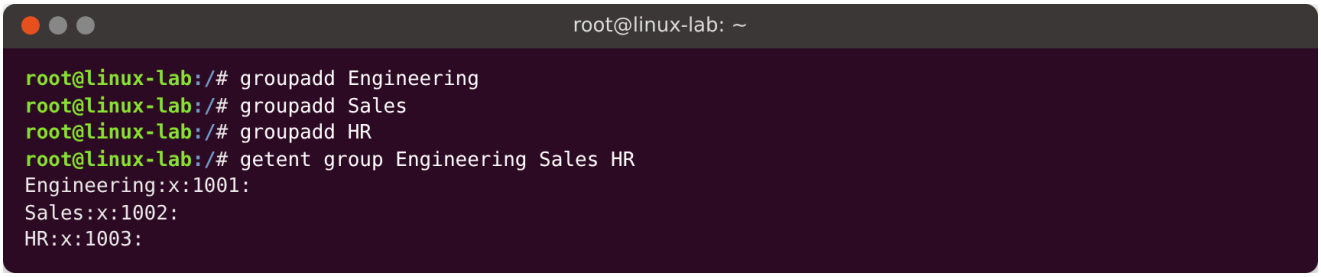
Task 2: Create a group for each department

How we did it: We create one group per department with `groupadd`. Each group has the same name as its department.

Code:

```
groupadd Engineering
groupadd Sales
groupadd HR
```

Verification: `getent group` shows that the groups exist in `/etc/group`.



Task 3: Create an administrative user for each department

How we did it: We create one admin per department with `useradd`:

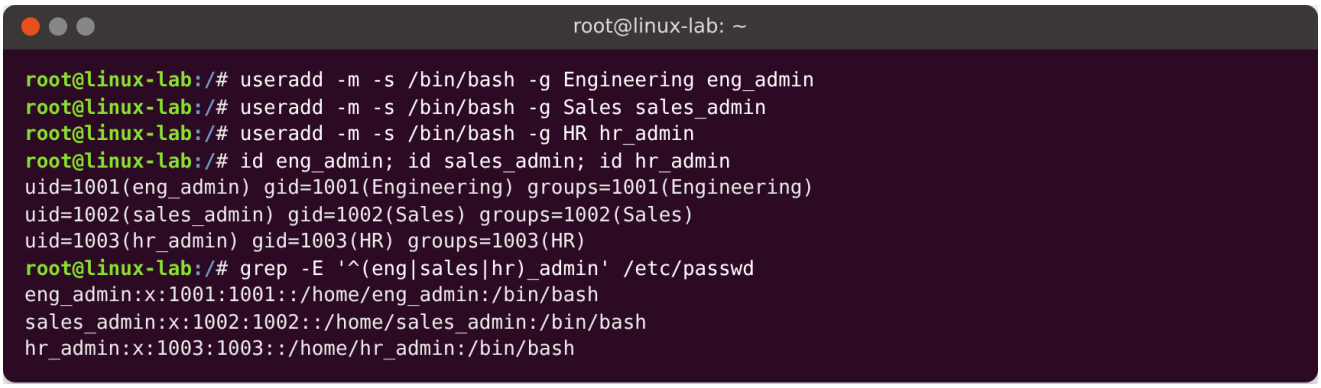
- `-m` creates a home folder for the user.
- `-s /bin/bash` gives the user a **Bash login shell** (3a).
- `-g <group>` makes the department group the user's **primary group** (3b).

Code:

```
useradd -m -s /bin/bash -g Engineering eng_admin
useradd -m -s /bin/bash -g Sales sales_admin
useradd -m -s /bin/bash -g HR hr_admin
```

Verification:

- `id` shows that the primary group (`gid`) is the department group.
- `/etc/passwd` shows that the login shell is `/bin/bash`.



Task 4: Create two additional users for each department

How we did it: We create two normal users per department, with the same options as the admins: a Bash login shell (4a) and the department as primary group (4b).

Code:

```
useradd -m -s /bin/bash -g Engineering eng_user1
useradd -m -s /bin/bash -g Engineering eng_user2
useradd -m -s /bin/bash -g Sales sales_user1
useradd -m -s /bin/bash -g Sales sales_user2
useradd -m -s /bin/bash -g HR hr_user1
useradd -m -s /bin/bash -g HR hr_user2
```

Verification: `id` shows the right primary group for every user, and `/etc/passwd` shows `/bin/bash` for all of them.

```
root@linux-lab: ~
root@linux-lab:/# useradd -m -s /bin/bash -g Engineering eng_user1
root@linux-lab:/# useradd -m -s /bin/bash -g Engineering eng_user2
root@linux-lab:/# useradd -m -s /bin/bash -g Sales sales_user1
root@linux-lab:/# useradd -m -s /bin/bash -g Sales sales_user2
root@linux-lab:/# useradd -m -s /bin/bash -g HR hr_user1
root@linux-lab:/# useradd -m -s /bin/bash -g HR hr_user2
root@linux-lab:/# for u in eng_user1 eng_user2 sales_user1 sales_user2 hr_user1 hr_user2; do id $u; done
uid=1004(eng_user1) gid=1001(Engineering) groups=1001(Engineering)
uid=1005(eng_user2) gid=1001(Engineering) groups=1001(Engineering)
uid=1006(sales_user1) gid=1002(Sales) groups=1002(Sales)
uid=1007(sales_user2) gid=1002(Sales) groups=1002(Sales)
uid=1008(hr_user1) gid=1003(HR) groups=1003(HR)
uid=1009(hr_user2) gid=1003(HR) groups=1003(HR)
root@linux-lab:/# grep -E '_user[12]' /etc/passwd
eng_user1:x:1004:1001::/home/eng_user1:/bin/bash
eng_user2:x:1005:1001::/home/eng_user2:/bin/bash
sales_user1:x:1006:1002::/home/sales_user1:/bin/bash
sales_user2:x:1007:1002::/home/sales_user2:/bin/bash
hr_user1:x:1008:1003::/home/hr_user1:/bin/bash
hr_user2:x:1009:1003::/home/hr_user2:/bin/bash
```

Task 5: Secure the department directories

5a – Owner and group

How we did it: `chown user:group` sets the owner and the group of a folder in one command. The department admin becomes the owner, and the department group becomes the group owner.

Code:

```
chown eng_admin:Engineering /Engineering
chown sales_admin:Sales /Sales
chown hr_admin:HR /HR
```

Verification: `ls -ld` shows the new owner and group.

```
root@linux-lab: ~
root@linux-lab:/# chown eng_admin:Engineering /Engineering
root@linux-lab:/# chown sales_admin:Sales /Sales
root@linux-lab:/# chown hr_admin:HR /HR
root@linux-lab:/# ls -ld /Engineering /Sales /HR
drwxrwxrwx 2 eng_admin   Engineering 4096 Oct  9 12:29 /Engineering
drwxrwxrwx 2 hr_admin    HR          4096 Oct  9 12:29 /HR
drwxrwxrwx 2 sales_admin Sales       4096 Oct  9 12:29 /Sales
```

5b–5e – Permissions

How we did it: One command covers all the permission rules: `chmod 1770`. Each digit means:

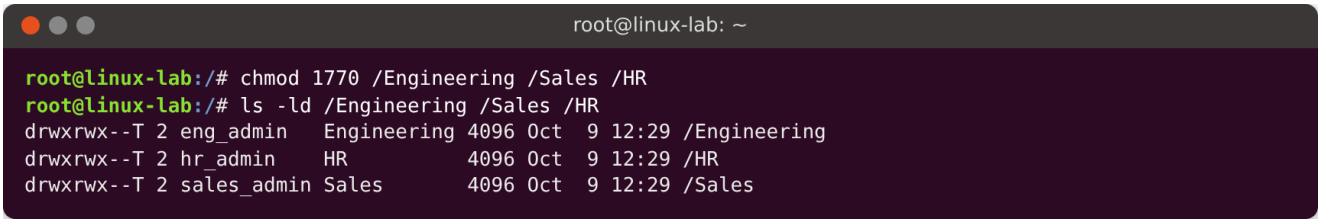
Digit	Meaning	Task
1	Sticky bit: only the owner of a file can delete it	5c
7	Owner (admin): read, write and execute, which is full access	5b
7	Group (department users): read, write and execute, which is full access	5d
0	Others: no permissions at all	5e

Code:

```
chmod 1770 /Engineering /Sales /HR
```

Verification: `ls -ld` shows `drwxrwx--T`:

- `rwx` for the owner,
- `rwx` for the group,
- `---` for others,
- `T` at the end, which means the sticky bit is on.



Test of task 5

How we tested it: We use `su - <user> -c '<command>'` to run a command as another user.

1. **eng_user1** creates a file in `/Engineering`. ✅ This works, so users have full access (5d).
2. **eng_user2** tries to delete eng_user1's file. ❌ *Operation not permitted*. The sticky bit works (5c).
3. **eng_user1** deletes their own file. ✅ This works.
4. **sales_user1** and **hr_admin** try to use folders from other departments. ❌ *Permission denied* (5e).



Task 6: Create a document in each department directory

How we did it:

1. `echo` writes the sentence into `confidential.txt` in each folder. The file has exactly one line (6b).
2. `chown` gives the file the same owner and group as its folder (6a).
3. `chmod 640` sets who can do what (6c):

Digit	Who	Permission
6	Owner (admin)	Read and write: only the admin can change the file
4	Group (department)	Read only
0	Others	No permissions

Code:

```
for d in Engineering Sales HR; do
  echo "This file contains confidential information for the department." > /$d/confidential.txt
done
chown eng_admin:Engineering /Engineering/confidential.txt
chown sales_admin:Sales /Sales/confidential.txt
chown hr_admin:HR /HR/confidential.txt
chmod 640 /Engineering/confidential.txt /Sales/confidential.txt /HR/confidential.txt
```

Verification: `ls -l` shows `-rw-r-----` and the right owner for every file, and `cat` shows the one line of text.

```
root@linux-lab: ~

root@linux-lab:/# echo "This file contains confidential information for the department." > /Engineering/confidential.txt
root@linux-lab:/# echo "This file contains confidential information for the department." > /Sales/confidential.txt
root@linux-lab:/# echo "This file contains confidential information for the department." > /HR/confidential.txt
root@linux-lab:/# chown eng_admin:Engineering /Engineering/confidential.txt
root@linux-lab:/# chown sales_admin:Sales /Sales/confidential.txt
root@linux-lab:/# chown hr_admin:HR /HR/confidential.txt
root@linux-lab:/# chmod 640 /Engineering/confidential.txt /Sales/confidential.txt /HR/confidential.txt
root@linux-lab:/# ls -l /Engineering /Sales /HR
/Engineering:
total 4
-rw-r----- 1 eng_admin Engineering 64 Oct  9 12:29 confidential.txt

/HR:
total 4
-rw-r----- 1 hr_admin HR 64 Oct  9 12:29 confidential.txt

/Sales:
total 4
-rw-r----- 1 sales_admin Sales 64 Oct  9 12:29 confidential.txt
root@linux-lab:/# cat /Engineering/confidential.txt
This file contains confidential information for the department.
```

Test of task 6

1. A department user **can read** the file. ✅
2. A department user **cannot change** it. ❌ *Permission denied*
3. The admin **can change** it. ✅
4. Users from other departments **cannot read** it. ❌ *Permission denied*

```
root@linux-lab: ~

root@linux-lab:/# su - eng_user1 -c 'cat /Engineering/confidential.txt'
This file contains confidential information for the department.
root@linux-lab:/# su - eng_user1 -c 'echo changed >> /Engineering/confidential.txt'
-bash: line 1: /Engineering/confidential.txt: Permission denied
root@linux-lab:/# su - eng_admin -c 'echo "This file contains confidential information for the department." > /Engineering/confidentia
l.txt && echo "eng_admin: file modified"'
eng_admin: file modified
root@linux-lab:/# su - sales_user1 -c 'cat /Engineering/confidential.txt'
cat: /Engineering/confidential.txt: Permission denied
root@linux-lab:/# su - hr_user2 -c 'cat /Sales/confidential.txt'
cat: /Sales/confidential.txt: Permission denied
root@linux-lab:/# su - sales_user2 -c 'cat /Sales/confidential.txt'
This file contains confidential information for the department.
```

Part 2 – User management script

How the script works

Our script is called `usermgmt.sh`. It asks the administrator for a group name, a username and a password, so **no names are hardcoded**. It follows a logical order:

- The group must exist before the user, because it becomes the user's primary group.
- The user must exist before we can give them a folder.

The script uses the special variable `$?`. It holds the result (exit code) of the last command: **0 means success**, and any other number means it failed. We use it to check whether a name already exists and whether each command worked.

Step by step:

- **Start:** The script checks that it is run as root. If not, it stops with an error.
- **(a) Create a new group:** The script asks for a group name. `getent group <name>` checks whether the group already exists. If `$?` is 0, the group exists, so the script shows an error and asks for another name. If the name is free, `groupadd` creates the group.
- **(b) Create a new user:** The same idea with `getent passwd <name>`. If the name is free, `useradd -m -s /bin/bash -g <group> <user>` creates the user with a Bash login shell and the new group as primary group.
- **(c) Create a password:** The password is typed twice with `read -s`, so it is hidden. If the two passwords are different or empty, the script asks again. `chpasswd` saves the password.
- **(d) Group membership:** `usermod -aG <group> <user>` adds the user as a member of the new group.
- **(e) Create a directory:** `mkdir /<username>` creates a folder at the root with the same name as the user.
- **(f) Ownership:** `chown <user>:<group> /<username>` gives the folder to the new user and group.
- **(g + h) Permissions:** `chmod 1770 /<username>` gives full control to the owner and the group, no access to others, and sets the sticky bit so only the owner of a file can delete it.
- **(i) Executable:** We make the script executable with `chmod +x usermgmt.sh` and run it with `./usermgmt.sh`.

At the end, the script prints a summary with `id`, `/etc/passwd` and `ls -ld`, which shows that everything worked.

The script

root@linux-lab: ~/lab

```
root@linux-lab:~/lab# cat usermgmt.sh
#!/bin/bash
# GIK2NV Linux Lab - Part 2
# Interactive user management script.
# Run as root: sudo ./usermgmt.sh

# The script must be run as root since it creates groups, users and directories
if [ "$(id -u)" -ne 0 ]; then
    echo "Error: this script must be run as root (use sudo)."
    exit 1
fi

# a. Create a new group - keep asking until a unique name is given
while true; do
    read -p "Enter a new group name: " groupname
    if [ -z "$groupname" ]; then
        echo "Error: group name cannot be empty. Try again."
        continue
    fi
    getent group "$groupname" > /dev/null
    if [ $? -eq 0 ]; then
        echo "Error: group '$groupname' already exists. Try another group name."
    else
        groupadd "$groupname"
        if [ $? -eq 0 ]; then
            echo "Group '$groupname' created."
            break
        else
            echo "Error: could not create group '$groupname'. Try another group name."
        fi
    fi
done

# b. Create a new user - keep asking until a unique name is given.
#    Bash login shell and the new group as primary group.
while true; do
    read -p "Enter a new username: " username
    if [ -z "$username" ]; then
        echo "Error: username cannot be empty. Try again."
        continue
    fi
    getent passwd "$username" > /dev/null
    if [ $? -eq 0 ]; then
        echo "Error: user '$username' already exists. Try another username."
    else
        useradd -m -s /bin/bash -g "$groupname" "$username"
        if [ $? -eq 0 ]; then
            echo "User '$username' created."
            break
        else
            echo "Error: could not create user '$username'. Try another username."
        fi
    fi
done
```

root@linux-lab: ~/lab

```
root@linux-lab:~/lab# cat usermgmt.sh
# c. Create a password for the user
while true; do
    read -s -p "Enter a password for $username: " password
    echo
    read -s -p "Confirm password: " password2
    echo
    if [ -z "$password" ]; then
        echo "Error: password cannot be empty. Try again."
    elif [ "$password" != "$password2" ]; then
        echo "Error: passwords do not match. Try again."
    else
        echo "$username:$password" | chpasswd
        if [ $? -eq 0 ]; then
            echo "Password set for '$username'."
            break
        else
            echo "Error: could not set password. Try again."
        fi
    fi
done

# d. Ensure the new user is a member of the new group
usermod -aG "$groupname" "$username"
if [ $? -eq 0 ]; then
    echo "User '$username' is a member of group '$groupname'."
fi

# e. Create a directory at the root (/) with the same name as the user
mkdir "/$username"
if [ $? -ne 0 ]; then
    echo "Error: could not create directory /$username."
    exit 1
fi
echo "Directory /$username created."

# f. Set ownership of the directory to the new user and group
chown "$username:$groupname" "/$username"

# g + h. 1770: owner rwx, group rwx, others none, sticky bit set so that
#        only the owner of a file can delete it from the directory
chmod 1770 "/$username"
echo "Ownership and permissions set on /$username."

echo
echo "===== Summary ====="
id "$username"
getent passwd "$username"
ls -ld "/$username"
```

Tests

Test 1 – Making the script executable (i)

Before `chmod +x`, the file is `-rw-r--r--`, so it cannot run. After `chmod +x`, it is `-rwxr-xr-x`, so it is executable.

root@linux-lab: ~

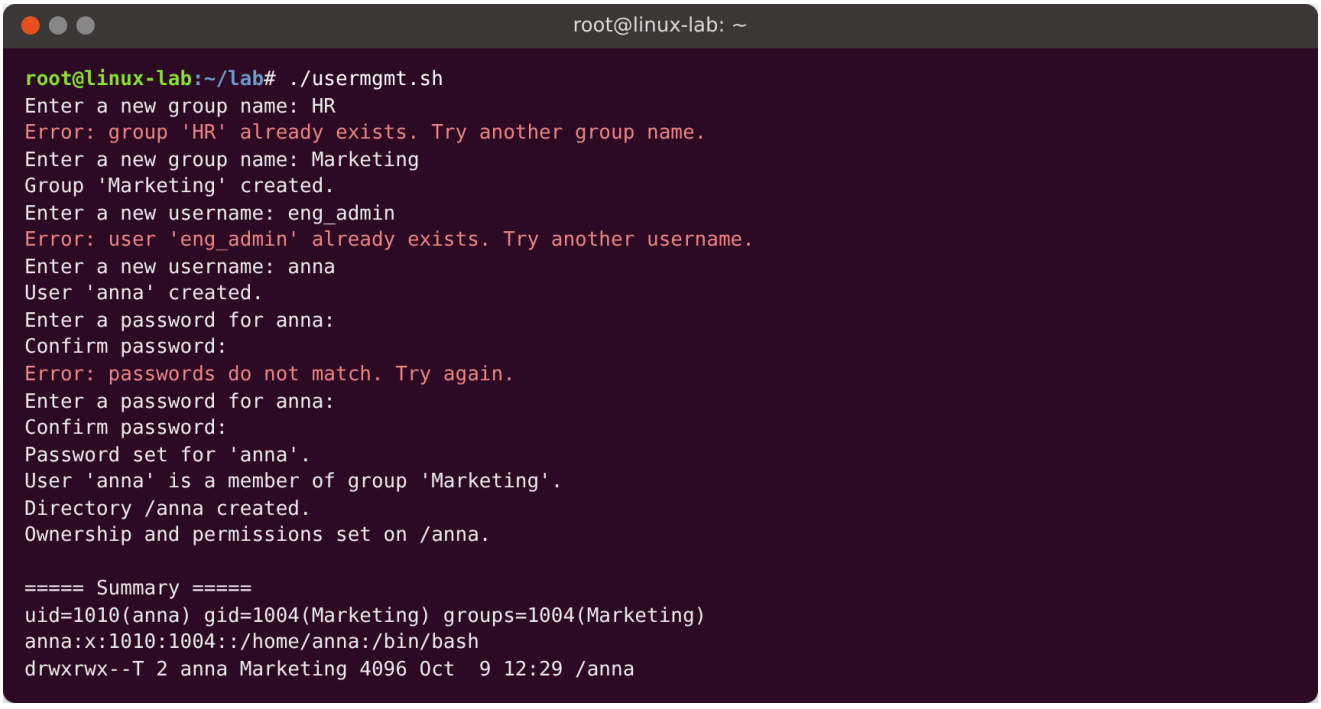
```
root@linux-lab:~/lab# ls -l usermgmt.sh
-rw-r--r-- 1 root root 3119 Oct  9 12:17 usermgmt.sh
root@linux-lab:~/lab# chmod +x usermgmt.sh
root@linux-lab:~/lab# ls -l usermgmt.sh
-rwxr-xr-x 1 root root 3119 Oct  9 12:17 usermgmt.sh
```

Test 2 – Wrong input and then correct input (a–h)

In this test we try every error case:

We typed	What happened	Task
`HR`	Error: the group already exists, so the script asks again	a
`Marketing`	The group is created	a
`eng_admin`	Error: the user already exists, so the script asks again	b
`anna`	The user is created	b
Two different passwords	Error: the passwords do not match, so the script asks again	c
The same password twice	The password is set	c

Then the script makes anna a member of Marketing (d), creates `/anna` (e), sets owner `anna:Marketing` (f) and sets the permissions `drwxrwx--T` (g, h).



Test 3 – Checking the result

1. `/etc/group` shows that **anna is a member of Marketing** (a, d).
2. `id` and `/etc/passwd` show that the **primary group is Marketing** and the **shell is /bin/bash** (b).
3. `/etc/shadow` shows that **a password is saved** (c).
4. `/anna` exists (e), is owned by **anna:Marketing** (f) and has **drwxrwx–T** (g, h).

5. anna **can create files** in her folder, and **other users are denied**.

```
root@linux-lab: ~
root@linux-lab:/# grep Marketing /etc/group
Marketing:x:1004:anna
root@linux-lab:/# id anna
uid=1010(anna) gid=1004(Marketing) groups=1004(Marketing)
root@linux-lab:/# grep anna /etc/passwd
anna:x:1010:1004::/home/anna:/bin/bash
root@linux-lab:/# grep anna /etc/shadow | cut -c1-30
anna:$y$j9T$y8x6894v.Yv5e9UCRz
root@linux-lab:/# ls -ld /anna
drwxrwx--T 2 anna Marketing 4096 Oct  9 12:29 /anna
root@linux-lab:/# su - anna -c 'touch /anna/anna_file && ls -l /anna'
total 0
-rw-r--r-- 1 anna Marketing 0 Oct  9 12:30 anna_file
root@linux-lab:/# su - eng_user1 -c 'ls /anna'
ls: cannot open directory '/anna': Permission denied
```

Test 4 – Running the script again with other names

This shows that the script works for **any names**. It also tests an empty group name, and it shows that `Marketing` and `anna` are now rejected because they already exist.

```
root@linux-lab: ~
root@linux-lab:~/lab# ./usermgmt.sh
Enter a new group name: Marketing
Error: group 'Marketing' already exists. Try another group name.
Enter a new group name:
Error: group name cannot be empty. Try again.
Enter a new group name: Support
Group 'Support' created.
Enter a new username: anna
Error: user 'anna' already exists. Try another username.
Enter a new username: bob
User 'bob' created.
Enter a password for bob:
Confirm password:
Password set for 'bob'.
User 'bob' is a member of group 'Support'.
Directory /bob created.
Ownership and permissions set on /bob.

===== Summary =====
uid=1011(bob) gid=1005(Support) groups=1005(Support)
bob:x:1011:1005::/home/bob:/bin/bash
drwxrwx--T 2 bob Support 4096 Oct  9 12:30 /bob
```

```
root@linux-lab: ~
root@linux-lab:/# id bob
uid=1011(bob) gid=1005(Support) groups=1005(Support)
root@linux-lab:/# ls -ld /bob
drwxrwx--T 2 bob Support 4096 Oct  9 12:30 /bob
root@linux-lab:/# ls -ld /anna /bob
drwxrwx--T 2 anna Marketing 4096 Oct  9 12:30 /anna
drwxrwx--T 2 bob  Support   4096 Oct  9 12:30 /bob
```

Conclusion

Part 1: Every department has its own folder, owned by its admin and its group. Everyone in the department has full access, but files can only be deleted by their owner (sticky bit). Users from other departments have no access. Each folder has a confidential file that the department can read but only the admin can change.

Part 2: Our script does the same work automatically for any group and user name. It checks for duplicates with `getent` and `$?`, asks again when something is wrong, and always creates a correctly set up user and folder.

All tests worked as expected. The video presentation shows every step live.